

Horse Racing Tournament Management System – Requirements

1. Authentication & Authorization

- Register account
- Login
- Logout
- JWT Authentication
- Role-based Authorization
- Password hashing
- Session/token management
- Manage profile
- Block/Activate account

2. User Roles

- Admin
- Horse Owner
- Jockey
- Race Referee
- Spectator
- Third-party Horse Verification Staff

3. Admin Requirements

- User Management
 - Create user
 - Update user
 - Delete user
 - Block/Activate user
 - Assign roles
 - Manage permissions
- Tournament Management
 - Create tournament
 - Update tournament
 - Delete tournament
 - Create rounds
 - Create races
 - Schedule races
 - Manage race status
- Registration Management
 - Approve horse registrations
 - Reject horse registrations
 - Manage approved participants

- Horse Management
 - Manage horse list
 - Remove invalid horses
 - View horse verification status
- Jockey Management
 - Manage jockey list
 - Suspend jockey
 - Approve jockey eligibility
- Referee Management
 - Assign referees to races
- Result Management
 - Publish official results
 - Update leaderboard
 - Calculate rewards
- Prediction Management
 - Configure prediction system
 - Configure reward rules
 - Open/Close prediction gate
 - View prediction statistics
- Dashboard & Reports
 - View statistics
 - Export reports PDF
 - View top horse
 - View top jockey
 - View tournament statistics

4. Horse Owner Requirements

- Register Horse Owner account
- Manage horses (CRUD)
- Upload horse images
- Register horse into tournament
- Search jockeys
- Send jockey invitations
- Manage invited jockeys
- Confirm jockey for race
- View horse race schedule
- Confirm horse participation
- View race details
- Track race results
- Track rewards and earnings

5. Jockey Requirements

- Register Jockey account
- View invitations
- Accept invitation
- Reject invitation
- View assigned races
- View assigned horse details
- View personal schedule
- Track personal performance
- View leaderboard

6. Race Referee Requirements

- Check horse before race
- Validate jockey and horse
- Monitor race progress
- Record race violations
- Process penalties
- Confirm raw race results
- Create race reports

7. Spectator Requirements

- View tournaments
- View race schedules
- Track live race results
- View leaderboard
- Participate in predictions
- View prediction history
- Claim prediction rewards
- Receive notifications

8. Third-party Horse Verification Requirements

- Verify horse information
- Validate horse documents
- Validate health certificates
- Upload verification documents
- Upload inspection reports
- Approve horse verification
- Reject horse verification
- View horse verification history

9. Horse Management Requirements

- Add horse
- Update horse
- Delete horse
- View horse details
- Store horse breed
- Store horse age
- Store horse weight
- Store horse gender

- Store horse health status
- Store horse racing history
- Store horse verification status

10. Tournament & Race Requirements

- Create tournaments
- Manage tournament rounds
- Create races
- Assign horses into races
- Assign jockeys
- Assign referees
- Manage race participants
- Manage race status
- Store race distance
- Store race time
- Store race location

11. Result & Leaderboard Requirements

- Store finish time
- Store ranking
- Store penalties
- Store rewards
- Update leaderboard automatically
- Calculate total points
- Calculate total wins

12. Prediction System Requirements

- Predict winning horse
- Predict top ranking
- Use betting points
- Store prediction results
- Calculate rewards
- Reward spectators

13. Notification Requirements

- Send invitation notifications
- Send race schedule notifications
- Send result notifications
- Send reward notifications
- Send approval/rejection notifications

14. Search & Filter Requirements

- Search by horse name
- Search by jockey name
- Search by tournament
- Search by race
- Search by owner
- Search by referee

15. Non-functional Requirements

- Responsive UI
- Secure authentication
- Fast performance
- Real-time updates
- Scalable architecture
- Input validation
- Error handling